

Traffic Simulator - Part 1

Carried out within the scope of the "Telecommunications Networks" course.

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1 Introduction

This work aims to simulate a Poisson arrival process and to verify, through simulation, the theoretical behaviour of event arrivals over time. Two approaches were implemented in C to analyse and compare the obtained results.

The source code developed for this lab work is available at: <https://github.com/mLeOnOr/RTel>.

2 Event-Driven Simulation

In this approach, the simulation is based on an event-driven model, where the system evolves from one arrival event to the next. In this case the time of the next event is calculated from the time of the previous one by adding a randomly generated inter-arrival interval. This makes it possible to simulate the process by moving directly between consecutive arrivals.

For a Poisson process with rate λ , the time between two consecutive events follows an exponential distribution. The inter-arrival time c is then generated using the inverse transform method:

$$c = -\frac{1}{\lambda} \ln(1 - u) \quad (1)$$

, where u is a random number uniformly distributed between 0 and 1.

After each inter-arrival time is generated, it is added to the previous event time to determine the instant of the next arrival. This cumulative process continues until a predefined number of events is reached (in this case 100 samples).

Knowing that, for a Poisson process, the average value of the inter-arrival time is given by:

$$E[c] = \frac{1}{\lambda} \quad (2)$$

, assuming $\lambda = 5\text{calls/seconds}$, the expected mean value is 0.2s.

After obtaining the sequence of inter-arrival times, a histogram was created to visualize their distribution, which represents how frequently c fall within specific ranges of values. In this sense, the range of possible values was divided into equal intervals of width Δ , which defines the step between consecutive bins of the histogram. In this implementation, $\Delta = \frac{1}{5\lambda}$, resulting in 25 intervals of 0.04s.

For each generated inter-arrival time, its corresponding histogram bin was determined by dividing the value of c by Δ and rounding down to the nearest integer, identifying which interval the value belongs to. The counter associated with that bin was incremented, building the histogram progressively as more samples were generated. Values greater than the maximum considered range were grouped into the last bin to ensure all data points were represented.

The simulation produced an average inter-arrival time of $E[c] \approx 20.55\text{ms}$, with the corresponding histogram presented in the following figure.

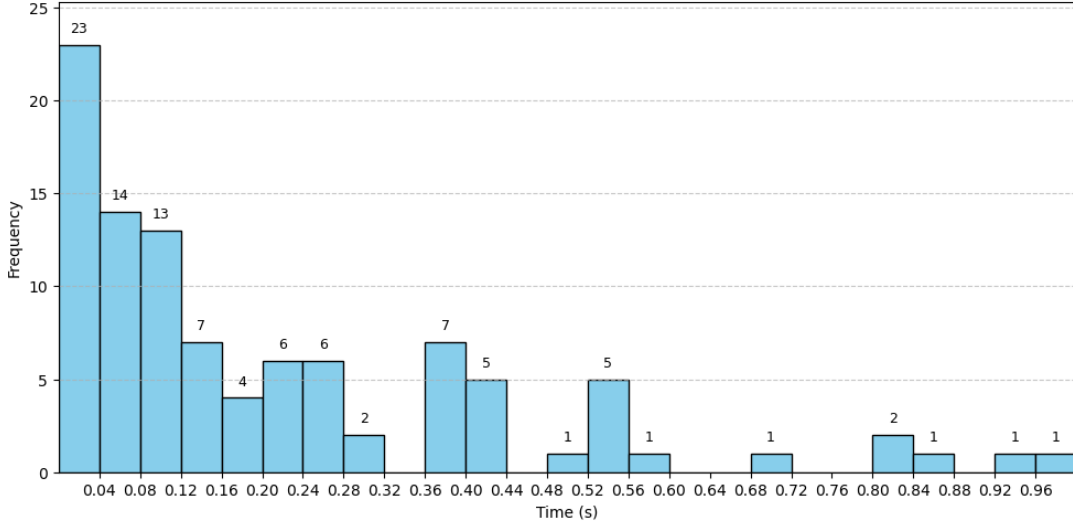


Figure 1: Histogram of inter-arrival times from event-driven simulation.

3 Time-Stepped Simulation

In this approach, the simulation is based on a time-stepped model, where time is divided into infinitesimal intervals of duration Δt , which was considered to be 1ms. At each step, it is checked whether an event occurred according to the probability associated with a Poisson process. This method follows the fundamental definition of a Poisson process, where the probability of one event occurring in a short time interval is: $P(\text{event in } \Delta t) = \lambda \Delta t$.

Similar to the previous approach, a random number u , uniformly distributed between 0 and 1, was generated at each iteration. In this case, if $u < \lambda \Delta t$ we could consider that an event have occurred during that interval. The current time value was then stored as the instant of the event, and the count of generated events increased by one. This process continues until the desired number of events is reached.

Unlike the event-driven approach, where the simulation jumps directly from one event to the next, this method examines every small time step, even when no event occurs. As a result, it requires significantly more iterations to produce the same number of arrivals, which increases the execution time.

After completing the simulation, the inter-arrival times between consecutive events were computed and used to calculate the mean value, resulting in $E[c] \approx 21.32\text{ms}$, very similar to the result obtain in the previous approach. A histogram of the results was also produced, using the same procedure mentioned before, with equally spaced intervals of width $\Delta = \frac{1}{5\lambda}$.

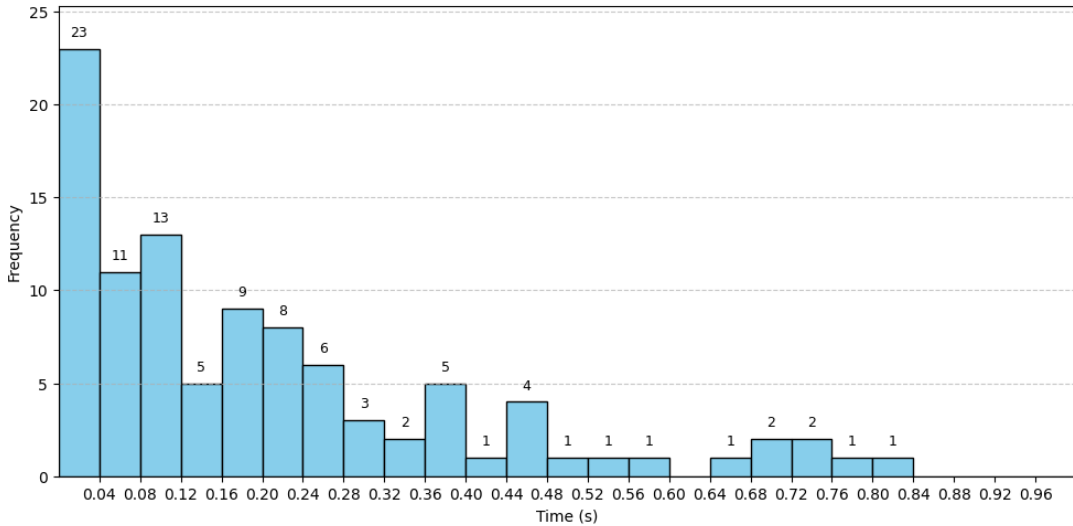


Figure 2: Histogram of inter-arrival times from time-stepped simulation.